Supplementary Information for: Forecasting against naive benchmarks: a scored test of process-based modules on Northern cod and the Edwards Aquifer

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Abstract

This Supplementary Information records material that supports the main text without carrying its argument. Part A (Northern cod): pass order and protocol status (S1); origin sets (S2); Diebold–Mariano and paired block-bootstrap grids with block-length sensitivity (S3); period decomposition, fixed windows and variant passes (S4); fitted parameters (S5); floor-forecast census (S6); secondary scores (S7); simulation design and deterministic primary-rule rerun, decomposition and carried-forward alternative-rule summaries (S8); lagged-start deficit (S9); outcome years (S10); vintage and threshold record (S11); 2022–2023 observation suspension (S12); runner reproducibility and verification (S13); and catch-timing sensitivity (S14). Part B (Edwards J-17): score-rule chronology and separate function-class dispositions (S15); origins and data processing (S16); fixed windows (S17); uncertainty implementations (S18); post-2007 threshold-indicator record (S19); Comal service series (S20); pumpage counterfactuals (S21); and climate-pass fixed windows with the Uvalde note (S22). Part C: baseline horizon profiles, full-sample autocorrelations and the exact rolling-origin J-17 decomposition (S23), plus this edition's verification record (S24). Main-text section, table and figure numbers are cited as such; SI sections are S1–S24. No archived forecast score was changed; the new checks and computations are the deterministic cod simulation rerun and the cross-system material in S23.

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Part A: Northern cod, NAFO 2J3KL

S1. Pass order and protocol status

The study is a fixed computational protocol with a recorded execution order, not a prospective registration: no dated pre-scoring protocol file exists for the empirical passes. The order below is established from the analysis code and its archived metadata. Passes 1 and 2 are labelled in the driver script `run_ladder.py` and distinguished in `results/meta.json` by the keys `catch_pass1` and `catch_pass2`; passes 3 and 4 are separate scripts that import from that driver, which fixes their order after it.

Three points bear on how the scores should be read. The five ladder modules were fixed before any pass was scored: the `SPECS` list in `run_ladder.py` defines M1, M1b, M2, M3 and M4 once, Specification B imports that list rather than redefining it, and none was altered after a score was seen. The scoring core, the fixed windows, the rolling-origin RMSE comparison and the persistence bar at both horizons, was coded before the first pass. Two elements of the retention rule were formalised after the pass-1 scores had been computed: the 5% tie band and the comparator declarations for the non-nested modules (M2 against M1, M4 against M3). Neither decides any retention outcome, the smallest pooled deficit against persistence being 17.1%. The capelin pass is the one place where a model was added after scoring began (the index multiplier), and it is reported as a variant, not as a member of the ladder. Specification B was run last and never pooled with Specification A; the constraint is recorded in the code's docstring and in `meta.json`.

S2. Origin sets and windows

The capelin acoustic index is observed in 1982 and 1985–2023 with gaps (1983, 1984, 1993–1995, 1997, 1998, 2006, 2016, 2020–2022 missing; 2023 = 331.3 kt). The multiplier pass carries the last observed value across gaps and drops origins with no carried value; the two-regime pass uses the 1991 capelin collapse as a declared break and allows r to differ before and after it, with the post-break value estimated only once post-break transitions are in the training window.

Table S1: Order in which the passes were run.

#	Pass	Data treatment	Script / archived output
1	Coarse-regime catch	240/120/5 kt (through 1991 / 1992 / from 1993)	`run_ladder.py` (pass 1); `fixed_window_scores.csv`, `rolling_summary.csv` (rows `catch = regime`), `rolling_forecasts_regime.csv`
2	Annual landings	Schijns et al. (2021) Table 1, t/1000	`run_ladder.py` (pass 2); same files, rows `catch = annual`; `rolling_forecasts.csv`
2b	Survey start	Forecast initiated at the survey-scaled state, annual catch	`run_survey_start()` within pass 2; `survey_start_forecasts.csv`
3	Capelin variants	Two-regime r at the 1991 break; acoustic-index multiplier on observed years	`run_capelin_regime.py`, `run_capelin_index.py`; `capelin_*_summary.csv`
4	Specification B	The ladder on the xteNCAM series alone, not pooled	`run_xte.py`; `xte_rolling_summary.csv`, `xte_fixed_window_scores.csv`
5	Uncertainty layer	DM statistics and block bootstrap on the archived per-origin files	`campaign_e1_dm_uncertainty_regime.py`; `e1_dm_uncertainty_regime.csv`
6	Original simulation	D1–D5 per `SPECIFICATION_v4.md` (locked 2026-09-10 before the code was written); D6–D7 per Amendment 1 (locked 2026-09-11)	`tools/sim_retention_power.py`; original archived CSV outputs
7	v68 deterministic rerun	Same D1–D7 and Protocol C; 200 replicates per cell, SHA-256-derived stable seeds, fail-loudly	`work/cod/sim_retention_power_repro.py` with `aggregate_power_simulation_repro.py`; aggregated tables, raw replicate CSV and hash-checked run manifest are written to `work/cod/` only after complete replicate coverage
8	Preceding edition	Re-execution of passes 1–6 in a third environment; paired bootstrap confirmation; period decomposition; rolling parameter record; catch-timing sensitivity	`paper11_forecasting_baselines_v67_src/work/cod/` in the repository

Table S2: Origin sets. All baselines are evaluated on the module's own origin set at each horizon. The mixed-origin persistence values that the archived Specification B summary also carries (87.6 and 317.7 kt on 63 and 59 origins) are not comparators and are not used.

Pass	Origins	n ($h = 1$)	n ($h = 5$)
Specification A, ladder (all catch passes, survey start, two-regime r)	1990–2014 ($h = 1$); 1990–2010 ($h = 5$); minimum eight training years	25	21
Specification A, capelin-index multiplier	1991–2014; 1991–2010 (origins with a carried acoustic value)	24	20
Specification B, ladder and two-regime r	1965–2023; 1965–2019; minimum twelve training years	59	55
Specification B, capelin-index multiplier	1988–2023; 1988–2019	36	32

S3. Uncertainty layer: full grids

Table S3 gives every module–benchmark pair of the archived uncertainty layer: the DM statistic (loss differences of squared errors; autocorrelation-consistent variance truncated at lag $h - 1$; reported as a z-score), the paired moving-block bootstrap interval of the RMSE gap (block $\max(h, 3)$, 20 000 replicates, seed 0) and its two-sided p-value. The bootstrap in the archived script draws block indices from a generator seeded identically for the two loss series, so the resample is paired; re-running it with an explicitly shared index draw reproduces every interval to the printed digit. Table S4 gives the block-length sensitivity (5 000 replicates per cell) for the M1 and M3 persistence margins.

The DM z-statistic is less stable than the bootstrap under the truncation lag: for the M1 persistence margin at $h = 1$ it moves from 1.14 at lag 0 to 2.30 at lag 3 on Specification A and from 2.81 to 1.70 on Specification B, crossing the conventional 1.96 in opposite directions, whereas the bootstrap p moves within 0.17–0.19 (A) and 0.036–0.049 (B) over blocks 3–9. Four of the 28 persistence and comparator rows have a bootstrap interval excluding zero with $|z| < 1.96$ (A, $h = 1$, M4 vs M3; B, $h = 1$, M3 vs persistence; B, $h = 5$, M1b vs persistence; B, $h = 5$, M4 vs M3). No verdict rests on either statistic where they disagree; the retention rule is applied to point estimates.

Table S3: Archived uncertainty layer: all pairs. RMSE in kt; gap = RMSE(A)−RMSE(B); CI is the 95% paired block-bootstrap interval of the gap; p is the two-sided bootstrap p-value. The M2-against-M1b rows are an alternative-comparator disclosure, not part of the rule.

Spec.	h	A	B	n	RMSE(A)	RMSE(B)	gap	DM z	95% CI	p
A	1	M1	persistence	25	120.5	98.0	+22.5	1.13	[-13.4, +52.2]	0.19
A	1	M1b	persistence	25	114.8	98.0	+16.8	0.98	[-19.4, +59.6]	0.38
A	1	M2	persistence	25	144.1	98.0	+46.0	1.29	[-12.3, +83.0]	0.45
A	1	M3	persistence	25	134.6	98.0	+36.6	0.99	[-16.8, +64.2]	0.93
A	1	M4	persistence	25	195.6	98.0	+97.5	1.53	[-9.1, +172.7]	0.28
A	1	M2	M1	25	144.1	120.5	+23.5	0.91	[-60.7, +66.9]	0.68
A	1	M4	M3	25	195.6	134.6	+60.9	1.21	[+4.4, +134.4]	< 0.005
A	1	M2	M1b	25	144.1	114.8	+29.3	0.91	[-67.6, +82.8]	0.70
A	5	M1	persistence	21	288.7	264.7	+24.0	1.84	[-43.1, +56.4]	0.28
A	5	M1b	persistence	21	288.6	264.7	+23.9	1.84	[-43.1, +56.4]	0.28
A	5	M2	persistence	21	398.2	264.7	+133.5	1.10	[-21.0, +217.5]	0.42
A	5	M3	persistence	21	366.4	264.7	+101.6	1.12	[-0.8, +151.4]	0.08
A	5	M4	persistence	21	488.2	264.7	+223.5	1.14	[-20.8, +409.4]	0.40
A	5	M2	M1	21	398.2	288.7	+109.5	0.98	[-77.1, +232.8]	0.81
A	5	M4	M3	21	488.2	366.4	+121.9	1.15	[-27.4, +263.2]	0.71
A	5	M2	M1b	21	398.2	288.6	+109.7	0.98	[-77.1, +232.8]	0.81
B	1	M1	persistence	59	119.5	84.4	+35.0	2.81	[+2.7, +70.8]	0.03
B	1	M1b	persistence	59	151.6	84.4	+67.2	3.60	[+18.3, +117.7]	0.01
B	1	M2	persistence	59	166.0	84.4	+81.6	3.22	[+36.4, +122.8]	< 0.005
B	1	M3	persistence	59	126.7	84.4	+42.3	1.85	[+1.0, +92.5]	0.04
B	1	M4	persistence	59	205.7	84.4	+121.3	3.20	[+53.7, +193.0]	< 0.005
B	1	M2	M1	59	166.0	119.5	+46.6	1.78	[-20.5, +104.5]	0.17
B	1	M4	M3	59	205.7	126.7	+79.0	3.07	[+35.6, +116.4]	< 0.005
B	1	M2	M1b	59	166.0	151.6	+14.4	0.51	[-67.5, +89.1]	0.77
B	5	M1	persistence	55	431.9	300.0	+131.9	2.06	[+33.2, +218.6]	0.01
B	5	M1b	persistence	55	445.5	300.0	+145.5	1.80	[+18.2, +250.9]	0.02
B	5	M2	persistence	55	1058.9	300.0	+758.9	2.15	[+363.6, +1038.5]	< 0.005
B	5	M3	persistence	55	930.1	300.0	+630.2	2.39	[+352.8, +845.6]	< 0.005
B	5	M4	persistence	55	1030.7	300.0	+730.8	2.35	[+407.3, +978.3]	< 0.005
B	5	M2	M1	55	1058.9	431.9	+627.0	1.97	[+195.7, +929.7]	< 0.005
B	5	M4	M3	55	1030.7	930.1	+100.6	1.88	[+20.2, +177.4]	0.01
B	5	M2	M1b	55	1058.9	445.5	+613.4	1.93	[+170.9, +945.1]	< 0.005

Table S4: Block-length sensitivity of the paired bootstrap interval (kt) for the persistence margins of M1 and M3. The archived layer uses block 3 at $h = 1$ and block 5 at $h = 5$. One classification changes with the block: the M3 one-step margin on Specification B includes zero for blocks of four or more.

Spec.	h	Module	gap	block 2	block 3	block 4	block 5	block 6	block 8
A	1	M1	+22.5	[-13, +49]	[-13, +52]	[-14, +54]	[-16, +55]	[-18, +56]	[-18, +56]
A	1	M3	+36.6	[-17, +77]	[-17, +65]	[-16, +56]	[-15, +54]	[-14, +55]	[-14, +55]
A	5	M1	+24.0	[-37, +56]	[-37, +56]	[-41, +58]	[-43, +56]	[-51, +55]	[-65, +56]
A	5	M3	+101.6	[-17, +181]	[-10, +179]	[-0.4, +167]	[-0.8, +151]	[-3, +158]	[-3, +152]
B	1	M1	+35.0	[+5, +67]	[+2, +71]	[+1, +74]	[+1, +74]	[+0.1, +75]	[+1, +73]
B	1	M3	+42.3	[+4, +87]	[+1, +95]	[-1, +95]	[-3, +93]	[-4, +90]	[-4, +81]
B	5	M1	+131.9	[+62, +198]	[+49, +208]	[+37, +215]	[+33, +221]	[+25, +224]	[+13, +232]
B	5	M3	+630.2	[+436, +803]	[+394, +830]	[+362, +848]	[+348, +843]	[+310, +850]	[+272, +849]

S4. Period decomposition: five-year horizon, annual pass, pooled margins

Table S5: Five-year rolling-origin RMSE (kt) by period of the forecast origin, with origin-matched persistence; superscripts are the number of origins at which the forecast sits at the numerical floor. Companion to Table 6.

Spec.	Origins	n	persist	M1	M1b	M2	M3	M4
A	1990–1995	6	479.4	501.1^{4}	500.8^{4}	740.0^{1}	674.5^{3}	909.2^{1}
A	1996–2012	15	78.6	127.6^{15}	127.6^{15}	54.7	77.1	54.9
A	1996–2014	15	78.6	127.6^{15}	127.6^{15}	54.7	77.1	54.9
B	1965–1990	26	398.8	573.4^{10}	595.2^{17}	1186.2^{7}	1178.6^{7}	1252.0^{7}
B	1991–1995	5	301.9	211.2^{4}	27.5^{5}	1490.1	834.3^{4}	1223.5^{3}
B	1996–2012	17	123.3	178.5^{17}	178.5^{17}	250.9	217.3	217.8
B	2013–2023	7	119.9	367.8^{5}	406.6^{7}	1358.2	1012.9	1157.9
B	1996–2023	24	122.3	249.0^{22}	266.1^{24}	763.3	576.8	651.7

Table S6: Specification A, annual-landings pass: rolling-origin RMSE (kt) by period at $h = 1$ (upper block) and $h = 5$ (lower block). The low-biomass band gives the same ordering as the coarse-regime pass.

Spec.	Origins	n	persist	M1	M1b	M2	M3	M4
A	1990–1995	6	193.6	209.6^{3}	187.1^{4}	325.6	312.1^{3}	419.1
A	1996–2012	17	19.6	63.0^{12}	69.9^{13}	17.2	15.0	19.8
A	2013–2014	2	66.7	127.2	134.7	33.5	28.7	40.2
A	1996–2014	19	28.5	72.5^{12}	79.2^{13}	19.6	17.0	22.9
A	1990–1995	6	479.4	501.1^{4}	500.8^{4}	732.3^{1}	647.5^{3}	906.6^{1}
A	1996–2012	15	78.6	127.6^{15}	127.6^{15}	50.9	73.0	49.7
A	1996–2014	15	78.6	127.6^{15}	127.6^{15}	50.9	73.0	49.7

Band-level intervals for the collapse bands (n = 5–6) are degenerate under a block bootstrap and are not reported. A leave-one-origin-out check on the Specification B collapse band confirms that the orderings there do not depend on a single origin: M1 and M1b are below persistence in all five leave-one-out sets and M3 in four of five, but these are the floor-forecast wins discussed in main-text Section 3.4.

Table S7: Post-1995 pooled margins against persistence (origins 1996 onward), with paired block-bootstrap intervals (block $\max(h, 3)$, 20 000 replicates). Specification A: 1996–2014 ($h = 1$) and 1996–2010 ($h = 5$); Specification B: 1996–2023 and 1996–2019. The bands were chosen after the data were seen; this table describes, it does not test.

Spec.	h	n	Module	RMSE	persist	gap	rel.	95% CI (kt)	p
A	1	19	M1	73.1	28.5	+44.6	+156%	[+28.9, +60.3]	< 0.005
A	1	19	M1b	79.8	28.5	+51.3	+180%	[+28.9, +69.5]	< 0.005
A	1	19	M2	19.4	28.5	-9.1	-32%	[-17.1, +6.6]	0.47
A	1	19	M3	16.7	28.5	-11.8	-41%	[-20.5, +3.3]	0.19
A	1	19	M4	22.1	28.5	-6.4	-22%	[-14.1, +9.0]	0.72
A	5	15	M1	127.6	78.6	+49.0	+62%	[+15.4, +68.1]	< 0.005
A	5	15	M1b	127.6	78.6	+49.0	+62%	[+15.4, +68.1]	< 0.005
A	5	15	M2	54.7	78.6	-23.8	-30%	[-27.5, +11.9]	0.76
A	5	15	M3	77.1	78.6	-1.5	-2%	[-2.7, +31.7]	0.19
A	5	15	M4	54.9	78.6	-23.6	-30%	[-27.2, +12.1]	0.78
B	1	28	M1	91.5	41.5	+50.0	+121%	[+26.3, +78.8]	< 0.005
B	1	28	M1b	124.0	41.5	+82.5	+199%	[+53.2, +117.7]	< 0.005
B	1	28	M2	102.7	41.5	+61.2	+148%	[-2.5, +93.3]	0.06
B	1	28	M3	62.3	41.5	+20.9	+50%	[-11.1, +47.0]	0.25
B	1	28	M4	87.4	41.5	+45.9	+111%	[-4.9, +73.3]	0.08
B	5	24	M1	249.0	122.3	+126.7	+104%	[+27.3, +196.8]	< 0.005
B	5	24	M1b	266.1	122.3	+143.8	+118%	[+27.3, +227.5]	< 0.005
B	5	24	M2	763.3	122.3	+641.0	+524%	[+97.9, +972.2]	< 0.005
B	5	24	M3	576.8	122.3	+454.5	+372%	[+46.6, +708.8]	< 0.005
B	5	24	M4	651.7	122.3	+529.4	+433%	[+61.7, +813.0]	< 0.005

S4.1 Fixed windows (moved from the main text of the single-system edition)

The fixed windows are diagnostic, not retention. From the 1983–1990 training window, in which SSB varied between 835 and 942 kt, every module forecasts continuation (Table S8, Fig. S1). The autonomous modules place the stock at the fitted equilibrium, 888.6 kt, and hold it there for five years; the catch-supplied modules, told that removals fall from 240 to 120 to 5 kt, forecast growth to about 1050 kt by 1993. The observed path falls from 862 to 9.7 kt. Persistence (670 kt RMSE) is the least bad because it at least does not grow; the autonomous modules (694) are close to it, and M2–M4 (819) are the worst because the catch cut, which in reality accompanied the collapse, is read by a stock-flow model as a reason for the stock to rise. On Specification B the pattern repeats from a 36-year training window (M1 817, M1b 906, M2 1898 kt against persistence 633).

The recovery window tells a different story. From 1995–2007, when SSB sat between 10 and 81 kt, M1 and M2 coincide (training-mean catch and realised catch are both about 5 kt) and over-predict the rebuild (120 kt RMSE

against 104 for persistence); M3 and M4 over-predict more (220 and 214), because the last training residual is a large positive increment (+28 kt in 2006–2007) that the AR term with $\phi = 0.95$ carries along the whole path. With annual landings the over-prediction is larger still (M3 609 kt), since the fitted K runs to its 5000-kt bound and production no longer saturates (Fig. S4 in S14). The one module that beats persistence in this window, M1b (90 and 78 kt), does so for a reason that undermines rather than supports it: its $\mathfrak{s}$ estimate descends to zero and its K to 106–130 kt, so it forecasts a flat line at a carrying capacity equal to the recent range. That is a persistence forecast produced by a production model whose parameters have lost their meaning (S5). On Specification B the recovery-stall window shows the same ordering: M1b 178, M1 254, M2–M4 about 268 kt, persistence 262.

Table S8: Fixed-window path RMSE (kt). Specification A, coarse-regime catch; the annual-landings pass and Specification B are given for the same windows. Persistence is frozen at the last training value; the training mean is the mean of the training window. Bold marks the lowest value in each column.

	Specification A			Specification B	
Forecaster	collapse 1991–95 coarse	recovery 2008–15 coarse	recovery 2008–15 annual	collapse 1990–95 annual	recovery stall 2013–24 annual
persistence	**669.8**	103.8	103.8	**632.7**	261.7
training mean	687.9	144.4	144.4	591.2	320.7
M1 autonomous	694.3	119.6	263.7	817.3	254.3
M1b depensatory	693.6	**89.8**	**78.1**	905.6	**178.0**
M2 stock-flow	819.4	119.6	302.8	1897.7	268.8
M3 AR residual	819.4	219.7	609.1	1767.8	267.7
M4 lagged start	819.1	214.1	586.3	1871.9	268.8

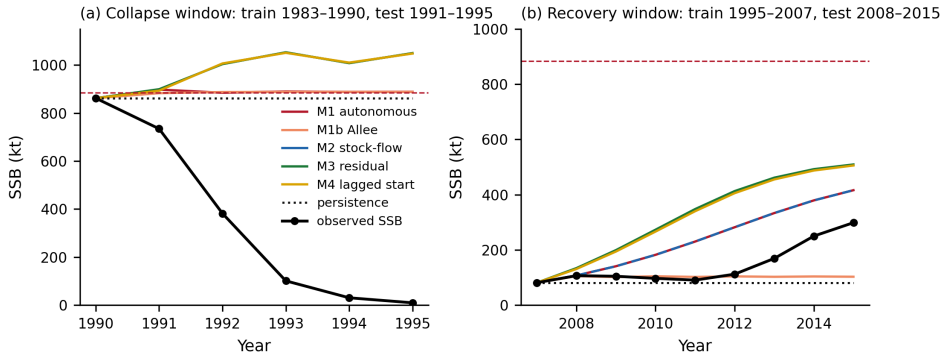


Figure S1: Fixed-window forecast paths on Specification A with the coarse-regime catch (the pass to which Table S8 refers). (a) Collapse window: every module forecasts continuation or growth from the 1990 state; M1b lies under M1. (b) Recovery window: M1 and M2 coincide (training-mean and realised catch are both close to 5 kt), M1b is a flat line at its fitted K, and M3 and M4 over-predict most because the last training residual, +28 kt in 2006–2007, is carried forward with $\phi = 0.95$.

S4.2 Variant passes

Four variant passes were added after the primary pass and declared in order (S1), each scored against persistence recomputed on its own origin set (Table S9). Replacing the coarse catch regime by annual landings on Specification A leaves the ordering unchanged. Starting the stock-flow forecast from a survey-scaled index value instead of the assessed state gives 128.1 and 331.0 kt. Allowing the intrinsic rate to differ before and after the 1991 capelin collapse, the mechanism most often proposed for the post-collapse productivity regime, gives 153.8 and 334.0 kt on A and 146.8 and 894.4 on B. Multiplying production by a capelin acoustic-index factor, which restricts the origins to the years with an acoustic estimate, gives 150.0 and 262.3 kt on A against 96.6 and 193.0 for persistence on the reduced set, and 132.0 and 491.7 on B against 78.6 and 287.9. No variant comes within the band on either horizon.

Table S9: Variant passes: rolling-origin RMSE (kt) of the variant module against persistence recomputed on the variant's origin set. n is the number of origins at $h = 1$ / $h = 5$.

		$h = 1$		$h = 5$	
Variant	Spec. (n)	module	persist	module	persist
Annual landings, M2	A (25/21)	160.4	98.0	393.8	264.7
Annual landings, M3	A (25/21)	153.6	98.0	351.5	264.7
Annual landings, M4	A (25/21)	206.3	98.0	486.4	264.7
Survey-start M2	A (25/21)	128.1	98.0	331.0	264.7
Two-regime r (1991 break)	A (25/21)	153.8	98.0	334.0	264.7
Two-regime r (1991 break)	B (59/55)	146.8	84.4	894.4	300.0
Capelin-index multiplier	A (24/20)	150.0	96.6	262.3	193.0
Capelin-index multiplier	B (36/32)	132.0	78.6	491.7	287.9

S5. Fitted parameters

Table S10: Fixed-window fits (one-step least squares on the training window). ϕ is the lag-one coefficient of the training residuals, clipped to $[-0.95, 0.95]$; σ their standard deviation (kt). M2–M4 share the stock-flow fit; M1 uses the training-mean catch, M1b the depensatory branch. Bounds: $r \in [10^{-3}, 2]$, $K \in [\max S + 10,\ 5000]$, $\mathfrak{s} \in [0, \max S]$; initialiser $r_0 = 0.3$, $K_0 = \max(1.5 \max S, 500)$.

Window (catch)	Module	r	K (kt)	$\mathfrak{s}$ (kt)	ϕ	σ
A collapse, train 1983–90 (coarse)	M1	1.935	1032.7	–	−0.14	33.8
	M1b	2.000	1058.9	0.0	−0.65	40.2
	M2–M4	1.935	1032.7	–	−0.14	33.8
A recovery, train 1995–2007 (coarse)	M1	0.458	500.0	–	0.95	11.8
	M1b	2.000	105.9	0.0	0.95	11.7
	M2–M4	0.458	500.0	–	0.95	11.8
A collapse (annual landings)	M1	1.942	1034.3	–	−0.14	33.8
	M1b	2.000	1062.0	0.0	−0.66	40.5
	M2–M4	1.991	1032.3	–	−0.02	47.0
A recovery (annual landings)	M1	0.370	5000.0	–	0.95	11.8
	M1b	2.000	129.8	0.0	0.95	11.6
	M2–M4	0.390	5000.0	–	0.95	10.6

The K estimate of the coarse-regime recovery fit, 500.0, is the initialiser $K_0 = \max(1.5 \max S, 500)$: the objective is flat in K above the observed range and the single L-BFGS-B start does not move it. Under annual landings the same fit runs to the 5000-kt upper bound. In both cases the production curve over the observed range (10–81 kt) is effectively linear with slope r (main-text Fig. 4).

Remark S1 (The compensatory branch is not the $\mathfrak{s} \to 0$ limit of the depensatory one). Substituting $\mathfrak{s} = 0$ in $a(S) = (S - \mathfrak{s})/(K - \mathfrak{s})$ gives $a(S) = S/K$,

Table S11: Specification A, coarse regime: parameters refitted at every rolling origin (expanding window from 1983). $\bar{C}$ is the training-mean catch used by M1 and M1b. From the 1992 origin, once the collapse transitions enter the training window, K sits at its bound for the stock-flow fit and the depensatory threshold moves to 585–734 kt; from 2011 it falls to about 5 kt.

	M2–M4 (stock-flow)			M1 (autonomous)			M1b (depensatory)		
Origin	r	K	ϕ	$\bar{C}$	r	K	r	K	$\mathfrak{s}$
1990	1.935	1033	-0.14	240.0	1.935	1033	2.00	1059	0.0
1991	1.557	1058	-0.03	240.0	1.557	1058	2.00	1042	0.0
1992	0.269	5000	+0.34	228.0	0.252	5000	2.00	1192	672.7
1993	0.250	5000	+0.67	207.7	0.212	5000	2.00	2110	635.0
1994	0.248	5000	+0.57	190.8	0.190	5000	2.00	3049	585.3
1995	0.248	5000	+0.56	176.5	0.170	5000	2.00	3300	597.9
1996	0.248	5000	+0.56	164.3	0.153	5000	2.00	3271	628.8
1997	0.248	5000	+0.56	153.7	0.138	5000	2.00	3332	649.2
1998	0.248	5000	+0.56	144.4	0.125	5000	2.00	3453	663.6
1999	0.248	5000	+0.56	136.2	0.114	5000	2.00	3633	673.3
2000	0.248	5000	+0.56	128.9	0.104	5000	2.00	3942	676.1
2001	0.248	5000	+0.56	122.4	0.095	5000	2.00	4198	681.7
2002	0.304	2537	+0.58	116.5	0.087	5000	2.00	4373	690.6
2003	0.247	5000	+0.56	111.2	0.080	5000	2.00	4476	701.6
2004	0.247	5000	+0.56	106.4	0.073	5000	2.00	4547	713.0
2005	0.247	5000	+0.56	102.0	0.067	5000	2.00	4606	723.7
2006	0.248	5000	+0.56	97.9	0.061	5000	2.00	4768	730.7
2007	0.248	5000	+0.56	94.2	0.057	5000	2.00	5000	734.1
2008	0.248	5000	+0.56	90.8	0.054	5000	1.57	5000	707.8
2009	0.248	5000	+0.56	87.6	0.050	5000	1.12	5000	656.9
2010	0.247	5000	+0.56	84.6	0.047	5000	0.80	5000	588.7
2011	0.246	5000	+0.56	81.9	0.044	5000	0.13	1411	5.3
2012	0.247	5000	+0.56	79.3	0.042	5000	0.12	1411	5.3
2013	0.247	5000	+0.56	76.9	0.083	1408	0.11	1411	5.2
2014	0.249	5000	+0.56	74.7	0.148	1114	0.11	1411	5.1

Table S12: Specification B: range of the refitted parameters over the origins of each band. The depensatory threshold stays at or near zero throughout; the stock-flow K is at or near its bound after 1993.

		M2–M4		M1			M1b		
Origins	n	r	K	$\bar{C}$	r	K	r	K	$\mathfrak{s}$
1965–1990	26	0.46–0.78	2482–5000	354–468	0.77–0.95	1845–2276	1.41–2.00	1642–2279	0.00–0.27
1991–1995	5	0.50–0.59	3355–4800	317–349	0.71–0.83	1869–1922	1.57–1.76	1654–1660	0.00–0.00
1996–2012	17	0.50–0.50	4802–5000	227–310	0.56–0.70	1789–1905	0.91–1.53	1651–2261	0.00–0.04
2013–2023	11	0.47–0.50	5000–5000	193–223	0.53–0.57	1682–1774	0.81–1.09	1603–2261	0.00–0.04

so $g(S) = rS(1 - S/K)(S/K)$, a cubic surplus, not the Schaefer form $a \equiv 1$. M1b at $\mathfrak{s} = 0$ is therefore a different model from M1, which is why the two differ slightly in the collapse window (693.6 against 694.3 kt) even though both thresholds are at zero; what M1b adds to M1 is a branch of the family,

not a limiting case of it.

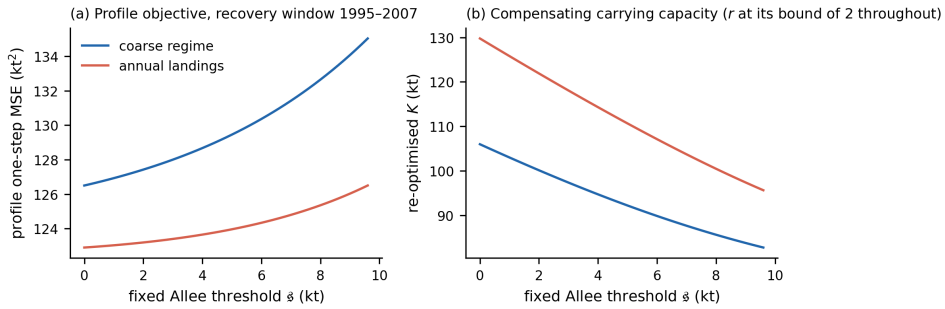


Figure S2: Profile of the one-step objective over a fixed depensation threshold $\mathfrak{s}$ in the recovery window (train 1995–2007), with r and K re-optimised at each value. (a) The objective rises monotonically from $\mathfrak{s} = 0$ and changes by 6.7% (coarse regime, 126.5 to 135.0 kt^2) and 2.9% (annual landings, 122.9 to 126.5) across the range below the smallest observed state (9.7 kt). (b) The compensating K falls as $\mathfrak{s}$ rises, with r at its bound of 2 throughout. The threshold is unidentified in this window; its least-squares value is the boundary.

S6. Floor-forecast census

A forecast is at the floor when the iterated map reaches the numerical floor $\epsilon = 10^{-3}$ kt; its absolute error then equals the observed SSB. Table S13 counts floor forecasts by band from the archived per-origin files (coarse regime for Specification A). The counts in the first block are those behind the superscripts of Table 6 and Table S5.

On Specification A the one-step M1 forecast is at the floor at every origin from 1993 to 2007; at 2008–2012 it forecasts 13–37 kt from states of 91–169 kt, and at 2013–2014 it forecasts 105 and 204 kt against 250 and 299. The mechanism is the constant catch: at the 2000 origin the training-mean catch is 129 kt (coarse) or 122 kt (annual) against a state of 34 kt, and at the 2010 origin 85 or 79 kt against 91 kt. On Specification B the training-mean catch is 317 kt at the 1995 origin and still 201 kt at the 2020 origin.

The log-RMSE diagnostic of the archived summaries uses $\log \max(\hat{S}, \epsilon)$ and is dominated by these floor forecasts; over the whole origin sets the floor binds for M1 at 15 of 25 ($h = 1$) and 19 of 21 ($h = 5$) origins on the Specification A annual pass and at 22 of 59 and 36 of 55 on Specification B, for M1b at 17/25, 19/21, 24/59 and 46/55, and for M2–M4 at between zero and eleven origins per cell. The raw RMSE, not the log column, is the retention score.

Table S13: Number of origins at which the forecast is at the floor, by band and horizon.

Spec.	Origins	n	M1	M1b	M2	M3	M4
A, $h = 1$	1990–1995	6	3	4	0	3	0
A, $h = 1$	1996–2012	17	12	13	0	0	0
A, $h = 1$	2013–2014	2	0	0	0	0	0
A, $h = 5$	1990–1995	6	4	4	1	3	1
A, $h = 5$	1996–2010	15	15	15	0	0	0
B, $h = 1$	1965–1990	26	2	3	0	0	0
B, $h = 1$	1991–1995	5	3	3	0	3	1
B, $h = 1$	1996–2012	17	17	17	0	0	0
B, $h = 1$	2013–2023	11	0	1	0	0	0
B, $h = 5$	1965–1990	26	10	17	7	7	7
B, $h = 5$	1991–1995	5	4	5	0	4	3
B, $h = 5$	1996–2012	17	17	17	0	0	0
B, $h = 5$	2013–2019	7	5	7	0	0	0

S7. Secondary scores

Table S14: Brier score of the hard below-threshold indicator implied by the point forecast ($\hat{S} <$ threshold against $S <$ threshold), origin-matched. Specification A uses the LRP of 884.6 kt, below which every origin and target after 1990 lies, so the indicator never varies and persistence scores zero trivially; Specification B uses $B_{\mathrm{lim}} = 276$ kt, which the series crosses in 1977, 1980, 1993 and 2016.

	A, $h = 1$	A, $h = 5$	B, $h = 1$	B, $h = 5$
persistence	0.000	0.000	0.068	0.291
M1	0.040	0.048	0.051	0.455
M1b	0.000	0.048	0.153	0.473
M2	0.080	0.143	0.068	0.364
M3	0.080	0.095	0.017	0.309
M4	0.080	0.143	0.068	0.382

On Specification B the M3 indicator has the lowest one-step Brier score (0.017 against 0.068 for persistence: one misclassified crossing instead of four), and M1's five-year score (0.455) is the worst because it forecasts the floor through two decades in which the stock sat just below B_{lim} and then crossed it. The direction-of-change score, the fraction of consecutive targets whose observed and forecast changes agree in sign, is 0.80–1.00 for the structural modules on Specification A and undefined for persistence (its forecast change is zero); it separates modules on an outcome that scarcely varies and carries no retention weight.

An interval-scale comparison is available on Specification B because the

assessment publishes 95% limits for its own predictand, with median relative width 0.44. Writing $D = |\hat{S}_{t+h|t} - S_{t+h}|/[(\mathrm{hi}_{t+h} - \mathrm{lo}_{t+h})/2]$, the proportion of hindcast errors not exceeding half the published width is 56% for persistence, 36% for M1 and 59% for M3 at $h = 1$ ($n = 59$), and 18%, 5% and 7% at $h = 5$ ($n = 55$). These are not coverage probabilities: the published interval expresses uncertainty about true biomass given the assessment's data, re-centring its width on a hindcast gives an interval with no established coverage, and the limits are right-skewed on the biomass scale (67 of 71 years asymmetric by more than 10%). The statistic indicates only that hindcast discrepancies are of the same order as, or larger than, reported target-state uncertainty, and that M3 reads marginally above persistence on it at one step where the RMSE ordering does not.

S8. Simulation: design and full decomposition

Table S15: Data-generating processes. All series have $T = 33$ years (the Specification A length) with Gaussian process noise $\sigma \in \{11.8, 33.8\}$ kt entering the state update; parameters are the archived fixed-window fits. Rolling origins use the eight-year minimum training length of Specification A. M3 and M4 were not simulated as truths.

DGP	Generating process	Source of parameters
D1	M1 autonomous Schaefer, $r = 1.935$, $K = 1032.7$, $C \equiv 240$, $S_0 = 900$	collapse-window coarse fit
D2	M1 autonomous Schaefer, $r = 0.458$, $K = 500$, $C \equiv 5$, $S_0 = 30$	recovery-window coarse fit
D3	M2 stock-flow, $r = 1.935$, $K = 1032.7$, catch 240/120/5 along the Specification A schedule, $S_0 = 900$	collapse fit and regime path
D4	M1b depensatory, $r = 0.458$, $K = 500$, $\mathfrak{s} = 15$, $C \equiv 5$, $S_0 = 30$	recovery fit with an identifiable threshold
D5	Persistence-true null, $S_{t+1} = S_t + \eta_t$, no production, $C \equiv 0$, $S_0 = 300$	–
D6	Drifting productivity, $r_t = 1.935\, e^{-0.05t} + 0.35$, $K = 1032.7$, catch 55% of the instantaneous $r_t K/4$	Amendment 1 (outside the class)
D7	Smooth path: noise-free state at $r = 0.9$, $K = 1032.7$, $C = 180$, scored series = state plus Gaussian noise	Amendment 1 (outside the class)

The design sheet (`SPECIFICATION_v4.md`) was locked at commit `d5d9758` on 2026-09-10 and the simulation code was written afterwards to implement it; no design element was changed after execution. Amendment 1 (D6, D7, and a $T = 71$ series-length cell for D1 and D5) was locked at

commit `bd0008f` on 2026-09-11; the $T = 71$ cell was started and abandoned as infeasible (45.6 s per rolling pass against 7 s at $T = 33$), so no series-length sensitivity is claimed. A module counts as retained in a replicate if it passes cod Protocol C1–C3, including the 5% relative-margin rule, at both horizons exactly as in the empirical rule. The completed v68 rerun covers 2,800 seeded DGP/noise series (14 cells × 200 replicates), scored against five modules, yielding 14,000 raw module rows. Replicate IDs 0–199 were processed in four disjoint 50-ID batches; seeds use the first eight bytes of SHA-256 over a fixed namespace, DGP cell, noise level to one decimal place, and replicate ID, with NumPy's PCG64 generator. All four batch manifests record source and output hashes; no replicate failed or produced a non-finite score. The estimates remain conditional on these fixed DGPs and parameter settings.

Table S16: Cod Protocol C operating characteristics. Panel A decomposes power loss averaged over the two process-noise levels: the persistence-only hurdle applies the 5% band at both horizons but omits the next-simpler comparator. Panel B gives each cell's observed rate and 95% Wilson interval; D1–D4 are truth-matching-module retention, while D5–D7 are any-module retention. Wilson intervals quantify Monte Carlo uncertainty only, not uncertainty in the fitted models or data-generating processes.

Panel A. Persistence-only versus full-rule power.

DGP	truth	persistence-only hurdle	full C rule	removed by comparator	conditional on hurdle
D1	M1	0.9625	0.9625	0.0000	0%
D2	M1	0.3850	0.3850	0.0000	0%
D3	M2	0.3300	0.1175	0.2125	64%
D4	M1b	0.1625	0.0075	0.1550	95%

Panel B. Cell-specific retention rate (count/200; rate [95% Wilson interval]).

DGP	$\sigma = 11.8$ kt	$\sigma = 33.8$ kt
D1 (M1 true)	195/200; 0.975 [0.943, 0.989]	190/200; 0.950 [0.910, 0.973]
D2 (M1 true)	132/200; 0.660 [0.592, 0.722]	22/200; 0.110 [0.074, 0.161]
D3 (M2 true)	21/200; 0.105 [0.070, 0.155]	26/200; 0.130 [0.090, 0.184]
D4 (M1b true)	0/200; 0.000 [0.000, 0.019]	3/200; 0.015 [0.005, 0.043]
D5 (persistence null)	5/200; 0.025 [0.011, 0.057]	15/200; 0.075 [0.046, 0.120]
D6 (drifting productivity)	144/200; 0.720 [0.654, 0.778]	154/200; 0.770 [0.707, 0.823]
D7 (smooth path + observation error)	194/200; 0.970 [0.936, 0.986]	190/200; 0.950 [0.910, 0.973]

Panel A separates the persistence hurdle from comparator attrition. Under D3, the persistence-only hurdle is passed in 12.0% of low-noise and 54.0% of high-noise series; the full rule retains true M2 in 10.5% and 13.0%. The comparator removes 3 of 24 and 82 of 108 persistence-hurdle survivors

Table S17: Descriptive rank-baseline check: proportion of 400 replicates (200 at each noise level) in which the generating module has the lowest one-step RMSE among the five modules; the nominal uniform-rank baseline is 0.20. Labels are descriptive comparisons, not formal tests of a random-ranking null.

DGP	true module ranks first	against chance
D1 (M1 true)	0.623	far above 0.20
D2 (M1 true)	0.740	far above 0.20
D3 (M2 true)	0.028	below 0.20
D4 (M1b true)	0.245	slightly above 0.20

(12.5% and 75.9%), respectively. Under D4 the persistence-only rates are 28.5% and 4.0%, but the full rule retains M1b in 0.0% and 1.5%; the comparator removes 57/57 and 5/8 survivors. Any module is retained under D3 in 105/200 and 122/200 series (95% Wilson intervals [0.456, 0.593] and [0.541, 0.675]), and under D4 in 61/200 and 11/200 ([0.245, 0.372] and [0.031, 0.096]), illustrating that low truth-specific power is not equivalent to no model being selected. Across D3's 400 simulated series, mean one-step RMSE is 85.5 kt for the true M2, versus 77.5 kt for M3 and 80.3 kt for M1b (78.2 kt for M4); estimation noise can therefore make the correct structure score worse than its siblings. Across D1–D4, non-truth modules are retained in 287 of 6,400 non-truth module–replicate pairs (4.5%), or 0.179 false module retentions per simulated series. Under the persistence-true null, 20 of 2,000 module–replicate pairs are retained (1.0%), averaging 0.050 false modules per series. Under D6, M1 is retained in 144/200 and 154/200 replicates at the two noise levels (298 M1 retentions among 300 module-retention events); under D7 the counts are 194/200 and 189/200 (383 of 394 events). A constant-productivity map thus out-predicts persistence on smoothly drifting or smoothly observed series often enough to clear both hurdles. Table S18 keeps the earlier alternative-rule summaries as reported, but its Protocol C reference row is recomputed from the completed v68 archive. The calculation code or complete formula for the alternative rules was not located for this edition; those alternatives are therefore not independently reproduced and do not inform the primary-rule conclusions.

The earlier archived analysis reported two candidate diagnostics for predicting from the data alone whether the rule has power: the dispersion of scores across modules (median coefficient of variation) correlated with true-module power at Spearman $\rho = 0.52$, while the best-module margin over persistence correlated at $\rho = 0.88$ (Pearson $r = 0.94$). These summaries are retained as historical reports, not independently reproduced for v68: their

calculation code and full cell-level formula were not located. The second diagnostic would in any case reuse the same out-of-sample scores consumed by the rule and cannot serve as an independent gate. Identifying a pre-score indicator of power remains open.

Table S18: Alternative decision-rule summaries. The Protocol C reference row uses the full v68 rerun; the other rules are carried forward from the earlier archived analysis and were not independently reproduced because their calculation code or complete formula was not located. Mean power is averaged over D1–D4 and both noise levels; specificity is one minus the any-module false-retention rate under D5. The earlier D6–D7 comparison is likewise historical and not recalculated here.

Decision rule	mean power	specificity
Protocol C as adopted (C1–C3, 5% band)	0.368	0.950
Without the comparator hurdle (C2, C3 only)	0.476	0.972
Persistence only, $h = 1$, 5% band	0.562	0.955
Persistence only, any margin	0.542	0.765
Full rule, no tie band	0.436	0.772
Full rule, 10% band	0.337	0.998
MASE < 1 against the naive benchmark	0.651	0.675
Information criterion, $n \log \mathrm{MSE} + 2k$	0.509	0.992

S9. Decomposition of the lagged-start deficit

M4 differs from M3 in two ways: it starts the path from S_{t-1}, and it therefore integrates one more catch year. A delayed-persistence control, $\hat{S}_{t+h} = S_{t-1}$, isolates the information delay. Its one-step RMSE is 184.4 kt on Specification A and 158.0 kt on Specification B, against 98.0 and 84.4 for ordinary persistence; the delay alone therefore costs 86.4 of M4's 97.5-kt deficit against persistence on A and 73.6 of 121.3 on B. The lagged start is an information-delay penalty, not a dynamical one, and M4 is not retained for the same reason that a one-year-stale persistence forecast would not be.

S10. The outcome years 2016–2021

Specification A ends in 2015. The continuation of the same assessment family is published in the RAM Legacy Stock Assessment Database (v4.66, stock COD2J3KL), and xteNCAM reads the same years independently (Table S19). Every year 2016–2021 lies above its 2015 origin in both vintages, so the window contains no event of the kind the collapse test probes, and persistence from the 2015 origin errs low throughout: MAE 129.2 kt and

RMSE 133.3 kt on the RAM vintage (107.5 kt MAE from the Table A2 origin of 298.65 kt), the same order as its scored one-step RMSE. The two vintages agree on the outcome years to within 1–26 kt (mean 16.8), while on the historical overlap they do not (S11). The restatement of the Edwards J-17 scores under Protocol C, which the single-system edition carried here, is now in main-text Section 4.3.

Table S19: Outcome years on the Specification A horizon. Persistence error is $|S_t - S_{2015}|$ from the RAM-vintage 2015 origin of 277 kt; the xteNCAM column is the revised model's reading of the same years.

Year	RAM v4.66 (kt)	xteNCAM (kt)	|difference|	persistence error (kt)
2016	340	339	1	63
2017	433	451	18	156
2018	394	369	25	117
2019	419	400	19	142
2020	440	414	26	163
2021	411	423	12	134

S11. Vintages and thresholds

Vintage overlap, 1983–2015. The NCAM Table A2 vintage (Specification A) and the RAM Legacy v4.66 continuation of the same assessment family (stock COD2J3KL) differ over 1983–2015 by +24.7 kt on average (Table A2 higher), by 26.8 kt in mean absolute value and by 131.2 kt at most, in 1984 (Fig. S3). The xteNCAM revision differs from Table A2 over the same years by 126.2 kt RMSE, with Table A2 higher by 58.5 kt on average. The 2015 origin is 298.65 kt on Table A2, 277 kt on RAM and 273 kt on xteNCAM.

Vintage-shifted LRP. The LRP defined as the 1983–1989 mean is 884.6 kt on the Table A2 vintage and 790.0 kt on the RAM vintage, a shift of 94.6 kt in the reference point from a revision of the same model family. Persistence from the 2015 origin under-predicts every outcome year 2016–2021 in both vintages: MAE 129.2 kt and RMSE 133.3 kt from the RAM origin of 277 kt, MAE 107.5 kt from the Table A2 origin. The outcome years agree across the RAM and xteNCAM vintages to within 1–26 kt (mean absolute difference 16.8 kt).

Threshold classification on one file. The xteNCAM file (1954–2024, 71 years) carries a ratio column that fixes B_{lim} at 276 kt (per-year implied values 250–289, mean 275.5). Classifying each year under $B_{\text{lim}} = 276$ kt

and under the 884.6-kt LRP gives 45 years at or above B_{lim}, 16 years at or above the LRP (1954–1969 only), and 29 disagreement years in two blocks: 1970–1992 without 1977–1979, and every outcome year 2016–2024. Under the 1980s-mean threshold the stock stands below the reference point for 55 consecutive years (1970–2024); under B_{lim} it is below for 26 years (1977–1979 and 1993–2015) and above from 2016 onward. The late-1970s episode (ratios 0.69–0.95) is a critical-zone reading, on the current model, of years the contemporary fishery regarded as healthy. The checkpoints 2005, 2017 and 2024 read 26, 451 and 342 kt. The October 2023 framework revision, which moved the stock's reported zone from critical to cautious without a change in the stock, is this experiment executed by the assessment authority. The scored RMSE comparison never uses the thresholds; every normative verdict layered on the forecasts, the Brier indicator included, inherits the dependence.

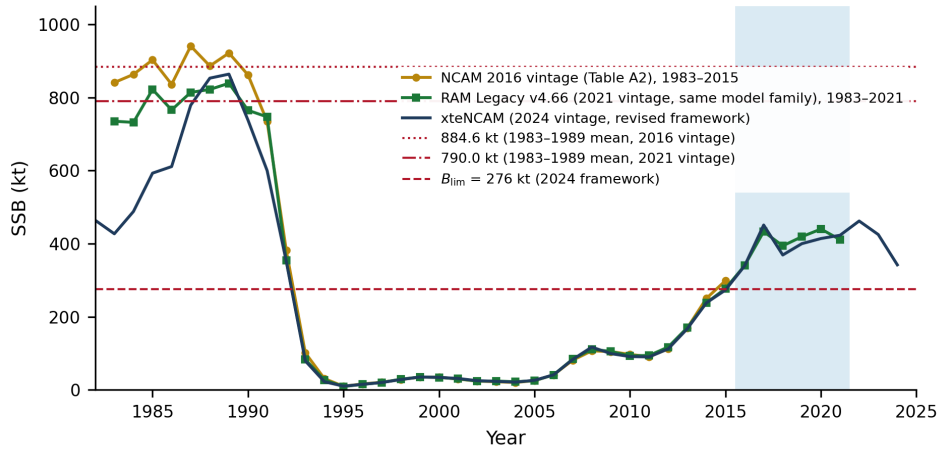


Figure S3: Three vintages of Northern cod SSB over 1983–2024: NCAM Table A2 (Specification A, 1983–2015), RAM Legacy v4.66 (the same assessment family continued to 2021) and xteNCAM Table 17 (Specification B). Horizontal lines mark the two Specification A LRP values (884.6 kt on Table A2, 790.0 kt on the RAM vintage) and $B_{\text{lim}} = 276$ kt. The outcome years 2016–2021 lie above the 2015 origin in every vintage and below 884.6 kt in every vintage.

S12. The 2022–2023 observation suspension

The test conditions on a complete annual series. The stock's real 2022–2023 history violates that condition. Partial coverage by the fall research-

vessel survey in 2021 and 2022 left the leading indicator of stock size uncomputable; no typical assessment update was possible in 2022 or 2023; the stewardship maximum was rolled over unchanged at 12,999 t; and the survey vessel that carries the series was condemned in February 2023. Table S20 assembles the episode from the assessment of record and the public record.

Table S20: The 2022–2023 observation suspension and its resolution (sources: Regular et al. 2025 and its Table 17 file; the 2024 reopening announcement; the 2025 and 2026 assessment coverage).

Period	Event
2021–22	Partial fall-survey coverage; leading indicator not computable
2022–23	No typical assessment update; stewardship maximum held at 12,999 t
Feb 2023	Survey vessel condemned
Mar 2024	Revised-framework assessment: SSB at 1.2 B_{lim} (95% CI 0.7–2.1; 22% critical-zone probability)
Jun 2024	Commercial fishery reopened at 18,000 t after 32 years
2025	TAC 38,000 t; reference point revised after being deemed overestimated
Apr 2026	Healthy-zone report: SSB $\approx$ 540 kt (CI 420–700); 70% probability above the upper stock reference

Read against a review-timing rule under which a review must land no later than the worst-case loss of the information it consumes, the episode is that rule's boundary case in the wild: the information loss ran past the review cadence, and the removals rule froze for two cycles while the observation map was dark. Abaee (2026, Periodic review as sampled governance) states the formal identity; the 2022–2023 suspension is its empirical instance, and it marks the practical limit of any scoring exercise that presumes the series exists.

S13. Reproducibility and verification record

Determinism. All computations are deterministic: within a fixed interpreter and library stack, repeated executions reproduce every result file byte for byte. An independent re-execution verified all 29 recorded output checksums and regenerated all 30 result files byte-identically; one archived file carries no pinned checksum and is covered by the regeneration comparison. Across environments (Python 3.12, numpy 2.1.3, scipy 1.14.1 against the recorded original environment), the intervention runners and the OLS-based prediction runners regenerate their outputs byte for byte, while the four optimiser-based forecast runners (L-BFGS-B fits) reproduce every scored

row at printed precision with one exception: the M1b rolling row on Specification B, whose depensatory optimum is environment-sensitive at the ±17 kt level ($h = 1$: 151.6 against 153.2 kt; $h = 5$: 445.5 against 462.5 kt). That sensitivity is the identification fragility of Section S5 and is immaterial to the verdict, which persistence wins at both horizons by margins far larger than the instability.

Third environment (preceding edition). For the preceding single-system edition the four forecast runners (`run_ladder.py`, `run_xte.py`, `run_capelin_regime.py`, `run_capelin_index.py`), the uncertainty script and the then-current simulation outputs were re-executed or re-read in a third environment (Python 3.13.16, numpy 2.3.5, scipy 1.17.1). Every result file, including the M1b rows, was reproduced at printed precision, and the persistence baselines were recomputed from the series on the archived origin sets and matched the archived origin-matched values (98.05, 264.72, 84.43, 299.98 kt). The paired block bootstrap was re-run from the archived per-origin files with an explicitly shared index draw and reproduced every interval of Table S3. The period decomposition, the rolling parameter record, the floor census and the catch-timing sensitivity were computed in the same environment from the archived files and the unmodified runner functions; their scripts are archived under `paperE1_cod_forecast_ladder_v69_src/work/`. The present v68 edition re-runs only the cod operating-characteristic simulation using the restartable v68 runner, four disjoint replicate-ID batches and the accompanying hash-checked aggregator; the complete 200-replicate-per-cell archive is described in SI S8. No empirical forecast runner was re-executed, and no archived cod forecasting score changed.

Verification script. `paper11_forecasting_baselines_v68_verification.py` (standard library only) regenerates every number of main-text Section 3.7, of S10 and of Sections S11–S12 from the archived data files (NCAM Table A2, RAM v4.66 time series, xteNCAM Table 17, the capelin index) and asserts the cross-checks, including the presence in the manuscript source of the pinned values and the suspension facts. All checks pass on the main text and this document taken together.

Scripts. (Run from the repository root; set `SRC` to the checked-out `paper11_forecasting_baselines_v68_src` directory.)

```
SRC="path/to/paper11_forecasting_baselines_v68_src"
python3 wave_e_cod/src/run_ladder.py
python3 wave_e_cod/src/run_xte.py
```

```
python3 wave_e_cod/src/run_capelin_regime.py
python3 wave_e_cod/src/run_capelin_index.py
python3 "campaign_e1_dm_uncertainty_regime.py"
# Historical pre-v68 simulation; not the source of Fig. 6 v68 rates.
python3 tools/sim_retention_power.py
# Current v68 operating-characteristic rerun: four disjoint 50-ID batches.
COD="$SRC/work/cod"
SIM="$COD/sim_retention_power_repro.py"
BATCH="$COD/power_batches"
AGG="$COD/aggregate_power_simulation_repro.py"
PLOT="$COD/plot_operating_characteristics.py"
for START in 0 50 100 150; do
  END=$((START + 50))
  NAME=$(printf 'rep_%03d_%03d' "$START" "$END")
  python3 "$SIM" --reps 50 --rep-start "$START" \
    --workers 4 --out-dir "$BATCH/$NAME"
done
python3 "$AGG" \
  --batches "$BATCH/rep_000_050" "$BATCH/rep_050_100" \
            "$BATCH/rep_100_150" "$BATCH/rep_150_200" \
  --reps 200 --out-dir "$COD"
python3 "$PLOT" \
  --summary "$COD/power_simulation_repro_summary.csv" \
  --output "$SRC/figs/fig6_operating"
python3 "$COD/paired_bootstrap.py"
python3 "$COD/post1995.py"
python3 "$COD/catch_timing_sensitivity.py"
python3 "$COD/make_figures.py"
```

S14. Catch-timing convention: a sensitivity outside the archived protocol

In `run_ladder.py` the estimator pairs the increment $S_{j+1} - S_j$ with the catch C_j of the same year, whereas the forecast path at origin t applies $C_{t+1}, \ldots, C_{t+h}$ to the steps $t \to t+1, \ldots, t+h-1 \to t+h$. The two conventions differ by one year. The estimation convention is the one under which SSB is a start-of-year quantity and C_t the removals during year t; the forecast convention gives the catch-supplied modules the catch of the target year, which under that reading occurs after the predictand is realised. The offset was noticed during the re-execution for this revision. Table S21 re-runs M2–M4 with the estimation convention in the forecast path (persistence and M1/M1b are unaffected). Under the coarse regime the two conventions differ only at the 1990–1992 origins, where the regime boundaries fall; under annual landings and on Specification B they differ at every origin.

Table S21: Rolling-origin RMSE (kt) of the catch-supplied modules under the archived forecast convention (C_{t+1} applied to the step $t \to t+1$) and under the estimation convention (C_t), with the ratio to origin-matched persistence in parentheses.

Spec. (catch)	Module	h	archived	estimation convention	persistence
A annual	M2	1	160.4 (1.64)	133.3 (1.36)	98.0
A annual	M2	5	393.8 (1.49)	390.8 (1.48)	264.7
A annual	M3	1	153.6 (1.57)	125.5 (1.28)	98.0
A annual	M3	5	351.5 (1.33)	352.2 (1.33)	264.7
A annual	M4	1	206.3 (2.10)	161.9 (1.65)	98.0
A annual	M4	5	486.4 (1.84)	438.7 (1.66)	264.7
A coarse	M2	1	144.1 (1.47)	113.4 (1.16)	98.0
A coarse	M2	5	398.2 (1.50)	367.8 (1.39)	264.7
A coarse	M3	1	134.6 (1.37)	105.2 (1.07)	98.0
A coarse	M3	5	366.4 (1.38)	341.5 (1.29)	264.7
A coarse	M4	1	195.6 (1.99)	138.9 (1.42)	98.0
A coarse	M4	5	488.2 (1.84)	409.8 (1.55)	264.7
B	M2	1	166.0 (1.97)	146.4 (1.73)	84.4
B	M2	5	1058.9 (3.53)	981.8 (3.27)	300.0
B	M3	1	126.7 (1.50)	94.9 (1.12)	84.4
B	M3	5	930.1 (3.10)	832.5 (2.78)	300.0
B	M4	1	205.7 (2.44)	130.3 (1.54)	84.4
B	M4	5	1030.7 (3.44)	898.3 (2.99)	300.0

Under the estimation convention every one-step deficit shrinks and no five-year deficit changes sign. M3 is 7% above persistence at one year on Specification A (coarse) and 12% on Specification B, still outside the 5% band, and 29% and 178% above at five years. The post-1995 bands of Table 6 are unchanged on Specification A (the catch is constant at 5 kt there under the coarse regime); on Specification B the post-1995 one-step M3 margin becomes $+10.4$ kt with interval $[-20.2, +46.8]$ ($p = 0.53$), M2 and M4 remain significantly worse ($+62.0$, $[+24.6, +98.7]$; $+45.9$, $[+11.8, +86.4]$), and all five-year margins remain far outside the band. The verdict of the main text does not depend on the convention; the size of the one-step deficits does, and a future application of the ladder should use the estimation convention throughout.

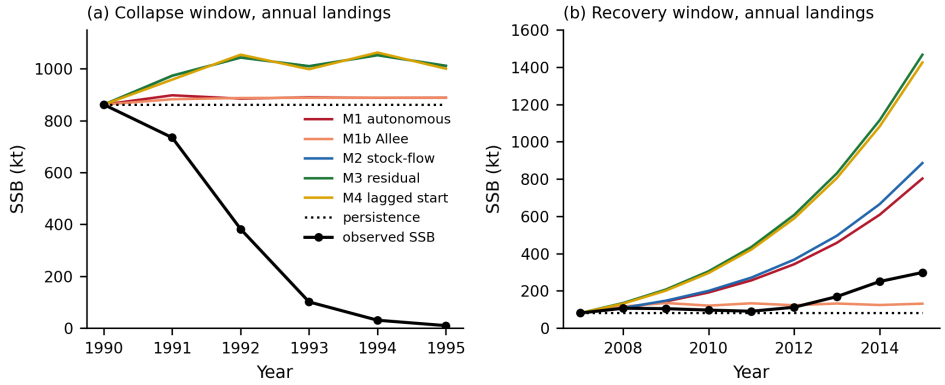


Figure S4: Fixed-window forecast paths on Specification A with annual landings (companion to Fig. S1 in S4). (a) Collapse window: as under the coarse regime, every module forecasts continuation or growth. (b) Recovery window: with K at its 5000-kt bound, production does not saturate and the catch-supplied modules over-predict more than under the coarse regime (M3 609 kt RMSE against 220); M1b is again a flat line at its fitted K.

Part B: Edwards Aquifer, index well J-17

S15. Protocol record: score gates, class dispositions and deviations

The scoring protocols for the primary pass and the climate pass were frozen and dated (2026-08-25) before the corresponding RMSE tables were computed, and are archived with the analysis code. The design is a fixed computational protocol rather than a prospective registration; the phrase "pre-registered" is avoided for that reason. The retention rule as frozen reads, verbatim:

> Let RMSE$(\cdot)$ denote the primary rolling-origin RMSE at a fixed horizon h. A causal module M is retained only if: (H1) RMSE(M) $<$ RMSE(persist), i.e. it beats last-value persistence, and (H2) RMSE(M) $<$ RMSE$(M_{\text{next-simpler causal}})$, i.e. it beats the next-simpler causal model. Retention is decided at $h = 1$ on the point-RMSE comparison; no tie tolerance and no minimum margin is imposed. Diagnostic oracles are excluded from retention. The Comal series is excluded from retention. Retention is decided on the head series only; the service series is scored after that decision is frozen.

The main text labels these hurdles (E1) and (E2). The primary score is RMSE of annual-mean J-17 in feet; secondary scores are mean absolute error and the Brier score of $\mathbf{1}\{\hat{H} < 660\}$, interpreted only for origins at or after 2007 (S19). A sign-hit rate of ΔH was listed in an earlier design iteration and is struck: no value is reported for it and it is not part of any retention decision.

Three distinctions in the frozen score rule, the broader class specification and the subsequent reporting are recorded here.

1. **M2m point score and class disposition.** M2m passes the Pass 1 H1/H2 inequalities at $h = 1$ (12.28 against persistence at 13.23 and M1 at 12.84 ft). The broader frozen specification classifies its climatological-flux map as the same affine-AR(1) function class; this is a class-count decision, not an H1/H2 failure or a post hoc change to the score gate (main-text Remark 1).

2. **Pass 2 comparator and class assessment.** The frozen Pass 2 protocol asks whether a causal recharge forecast lowers primary J-17 RMSE "relative to persistence and relative to M1"; M1 is therefore the declared comparator. Against persistence (13.23 ft) and M1 (12.84 ft), Renso (12.82), Rprecip (12.80) and combo (12.71 ft) pass the strict H1/H2 point gate, while Rar (13.25 ft) fails. The archived Pass 2 result metadata separately records the three point-gate passes as class-demoted M2m-plus-intercept variants and leaves `retained_as_structure` empty. If M2m (12.28 ft) were substituted as the comparator, none of the four climate variants would pass the strict point gate; that changes the score-gate count, not the separate structural disposition. The H1/H2 result and class assessment are reported separately; neither the uncertainty layer nor the $h = 5$ scores retroactively alters the frozen point gate.

3. **The struck sign-hit score.** A sign-hit rate for ΔH appeared in an earlier design iteration, was removed before scoring, and is not reported; it is not part of either score gate.

Post-freeze objects, labelled as such: the climate-pass fixed-window scores (S22), the pumpage counterfactuals (S21), the Comal service-series scoring (S20), and the uncertainty layer (S18) together with its independent replication. None replaces or alters a frozen verdict.

S16. Origin sets, windows and data processing

Rolling origin. Minimum 15 training years; the first origin is 1948 and forecasts target years through 2023, giving $n = 75$ origins at $h = 1$ and $n = 71$ at $h = 5$. Every model in the rolling tables, including M2m and the climate variants, is scored on the identical origin sets. Early rolling origins with the 15-year floor give 14 transitions for a four-parameter fit, so the causal family's rolling scores carry unstable early-window coefficients. Head forecasts are clipped to $[610, 710]$ ft before scoring. For example, the 1992-origin M2 one-step candidate is 719.1027 ft before clipping and is stored/scored as 710 ft; this is a model-domain bound, not an observed-data constraint.

Fixed windows. (1) Drought-of-record drawdown: train 1934–1950 (16 transitions), test 1951–1956. (2) Drought-of-record recovery: train 1934–1956 (22), test 1957–1961. (3) Pre-permit wet interval: train 1980–1990 (10), test 1991–1995. (4) Critical-period era: train 1997–2014 (17), test 2015–2023.

Head series. Annual means use all available daily highs of TWDB well 6837203. Years with fewer than 240 observations would be dropped; no year falls below the floor (minimum $n = 242$, 1939), so the rule never binds. 1935 ($n = 258$) and 1939 ($n = 242$) are retained as incomplete-coverage means; missing days are not interpolated; the published pre-continuous composite (Beverly Lodges extension) is used as issued; the complete estimation panel ends in 2023, whose values carry provisional TWDB status. The predictand is the calendar-year mean of daily *highs*; a daily-mean convention would sit slightly lower and matters near the 660/618-ft lines, and the Authority's Stage I rule is a 10-day mean, so the annual-mean indicator is a coarse proxy flagged where used. Stage I at 660 ft is a 2007 institutional rule and is not applied as if it existed in 1956.

Fluxes and climate. Recharge R is USGS total San Antonio-area recharge (10^3 acre-ft yr^{-1}), Puente stream-loss method, constructed independently of J-17 (Umphres and Choi, 2025). Pumpage P is well discharge from Edwards Aquifer Authority Table 1 (1934 onward). Total spring discharge is stored and not used as a driver; eastern-basin recharge was stored and not used. Climate predictors: September–November Niño 3.4 (HadISST, anomaly relative to 1991–2020) and calendar-year precipitation in Texas climate divisions 06 and 07 (NCEI nClimDiv). December–February precipitation that includes January of $t+1$ is not used. The registered twenty-column analysis panel (`wave_e_edwards/data/annual_panel.csv`) is the dataset of record

for all scored analyses; the nClimDiv raw file is not redistributed (provenance URL archived), so its three columns rebuild from the provenance source rather than from the archive, while the two Niño columns rebuild from the registered file to machine precision.

Information set. Information known at 31 December of year t comprises (H, R, P) through t, the September–November Niño 3.4 anomaly of year t and calendar-year precipitation of year t. A two-pool exchange module was specified and not fitted; no barrier or exchange term was fitted. Post-1997 pumpage no longer spikes with drought as in 1956 (321×10^3 acre-ft from wells); that change is already in P_t, and a separate critical-period switch adds no degree of freedom beyond the pumpage series.

S17. Fixed windows

Table S22 and Fig. S5 give the fixed-window scores. They are diagnostic, not retention. On the drawdown window the bolded M2 is a continuing-drought artefact, not a forecasting merit: the last observed R is already low, and the map trained on 1934–1950 has the wrong sign on pumpage ($\gamma = +0.021$), since pumping rose as the drought deepened and least squares aliases the human response into the state transition (simultaneity bias, compounded by the short window). That identification failure is why M2 "beats" the oracle there. The train-mean baseline is the best non-oracle forecast on the recovery window (14.07 ft) and the critical-period era (14.77 ft), consistent with its rolling five-year win; the residual-persistence rungs M3/M4 track M2 on the drawdown (18.12/18.23 against 18.11 ft); M4 is the best causal model on the pre-permit wet window (15.26 ft); and both fail with the causal family on the recovery and critical-period windows.

The recovery window is also where the $[610, 710]$ output clip binds on forecast paths. With persisted $(R, P) = (43.7, 321.1)$, the M2 pre-clip proposals are $617.1 \to 611.6 \to 606.8 \to 605.3 \to 605.3$ ft; the emitted values after clipping are $617.1 \to 611.6 \to 610 \to 610 \to 610$ ft, so the 610-ft floor first binds in 1959. The clip does not bind on the observed record (annual means 623–692 ft); it binds on persisted-drought forecast paths. The upper cap also binds in the rolling record: at origin 1992 and $h = 1$, M2's raw proposal is 719.10 ft and the saved forecast is clipped to 710 ft (SI S16). The audit script recomputes both fixed-window M2 RMSEs (18.11 and 55.32 ft) exactly from the registered panel. The 1957–61 recovery is a recharge pulse ($R_{1956} = 43.7$, $R_{1957} = 1142.6 \times 10^3$ acre-ft): persistence stays at the 1956 floor, M2 persists drought recharge and falls further, the oracle, given the

1957 flood year, tracks the rise. Recoveries on this specification are recharge events, not autonomous mean reversion, although the training mean (mean reversion) scores 14.07 ft, second only to the oracle, in this institutionally bounded, rapidly recharged system.

Table S22: J-17 fixed-window RMSE (ft). Bold marks the lowest RMSE of the window including the diagnostic oracle.

Window	persist	mean	M1	M2	M2m	M3	M4	oracle
Drawdown 1951–56	23.75	35.19	30.94	**18.11**	27.44	18.12	18.23	19.69
Recovery 1957–61	43.62	14.07	56.24	55.32	37.74	55.28	55.12	**12.26**
Pre-permit wet 1991–95	30.13	18.24	20.02	16.67	23.47	16.41	15.26	**7.18**
Critical-period era 2015–23	27.41	14.77	15.62	23.37	14.79	22.84	22.17	**8.70**

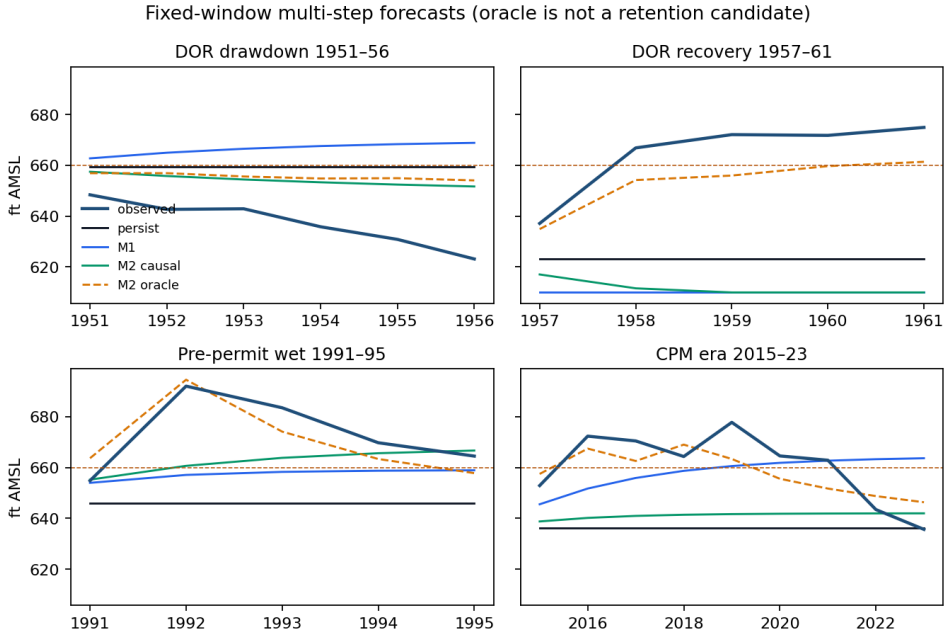


Figure S5: J-17 multi-step forecasts from the end of each training window. The oracle (dashed) is diagnostic and uses realised future R, P.

S18. Uncertainty layer: two implementations

The post-freeze uncertainty layer (`e3_audit_uncertainty.py`; seed 20260905) attaches Diebold–Mariano tests (Newey–West HAC, lag $h-1$)

and moving-block bootstrap intervals (block length 8, 10 000 replications) to every load-bearing margin, computed from the archived per-origin forecast files; main-text Table 8 reproduces its output. An independent replication (`campaign_e3_dm_uncertainty.py`, archived with its output `e3_dm_uncertainty.csv`) is a second implementation reading the same registered per-origin forecast file: HAC lag $h-1$ with unweighted truncation and population-variance scaling, block length h, 20 000 replications, seed 0, deterministic (byte-identical on re-execution). Its verified output reproduces every load-bearing reading: the M1 retention margin has DM $z = -0.86$ with block interval $[-1.22, +0.56]$ ft (bootstrap $p = 0.38$); the M2m edge over persistence has $p = 0.001$; the $h = 5$ climatology win has $p = 0.035$. It also registers two rows the main table does not carry, the $h = 1$ climatology loss (+2.94 ft against persistence, $p = 0.046$) and the M2 $h = 5$ loss (+12.4 ft, $p < 0.001$), both consistent with the readings above. The two implementations use different seeds, block lengths, HAC weighting and variance conventions, and agree on every conclusion: the 0.39-ft AR(1) retention margin is statistically indistinguishable from zero, the M2m edge is the only one-year margin that separates from noise, and the five-year climatology win survives.

S19. Threshold indicator scores on the post-2007 origins

The second secondary score is scoped to the post-2007 origins, where the 660-ft Stage I rule is in force. On those origins ($n = 16$ at $h = 1$) the committed records give 660-ft Brier scores of 0.31 (persistence), 0.25 (M1), 0.25 (M2m), 0.19 (oracle), with the one-year RMSE ranking unchanged (persistence 13.09, M1 12.16, M2m 12.18, M2 13.31, oracle 8.03 ft); at $h = 5$ ($n = 12$) the M1–persistence ordering reverses on the subsample (17.16 against 25.10 ft) and the training mean scores 16.41 ft, reported without changing the one-year retention statement. For deterministic 0/1 forecasts the Brier score is a misclassification rate of a coarse annual-mean proxy for the Authority's 10-day rule on a small subsample; the full record is archived as `wave_e_edwards/results/rolling_modern_2007.csv`.

S20. The service series after the retention freeze

The map $Q = c_0 + c_1 H$ fitted on 1934–1950 ($c_0 = -2876$, $c_1 = 4.77$) is a linear channel of the same state indexed by J-17; the full-sample contemporaneous correlation is $r = 0.986$ (Fig. S6). The channel fails in the drought tail, and the direction matters: at the 1956 annual mean (623.15 ft)

it reads ≈ 97 cfs against an observed ≈ 32 cfs annual mean, so it predicts non-cessation, and its zero-discharge level ($2876/4.77 \approx 602.9$ ft) lies below every observed head (daily minimum 612.5 ft), so the map never predicts cessation where cessation occurred. Near-linear at ordinary heads, the channel fails at cessation exactly where management cares. One-year Comal RMSE (cfs), scoring the same ladder directly on the Comal series over the same rolling origins ($n = 75$): persist 71.9, M1 69.0, M2m 68.7, M2 74.8, M3 73.8, M4 73.4, train-mean 89.7, oracle 45.3; the ranking mirrors the head ranking, as it must for a linear channel. The service series does not change retention, and it cannot: Comal is an independent measurement (a USGS spring gauge, not used to construct J-17) but not an independent information source.

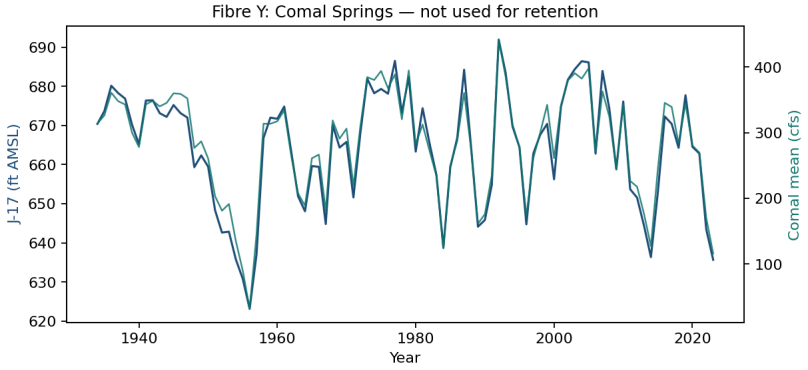


Figure S6: Comal annual mean against J-17. Contemporaneous $r = 0.986$. The service series is a measured rating curve of the same state.

S21. Pumpage counterfactuals

A declared scenario layer passes counterfactual pumpage paths through the pre-permit affine map (train 1980–1990, the ladder's window 3; the short-window identification caveat applies to the map itself), from the observed 1990 head (645.8 ft) over 1991–2023 with the actual recharge sequence (Table S23). The scenarios are counterfactual simulations of the fitted map, not forecasts, and carry no retention implication. Four readings follow. First, pumpage is a secondary lever in this map: the full spread of counterfactual policies spans 630.9–646.8 ft at 2023, and the map's own RMSE against the observed record (7.2–14.2 ft) is of the same order as the policy spread, so the 5-ft effects are within model error and are not policy estimates. Second, the 20%-cut path is the closest to the observed record of the four scenarios (7.19

against 8.56 ft), an artefact of the map running high on this window (the actual-pumpage path ends 5.9 ft above the observed 2023 head of 635.7 ft), not evidence that the Authority's pumpage was near-optimal. Third, the direction is right and the magnitude modest: the 20% cut raises the simulated 2023 head by 5.2 ft relative to the actual-pumpage path. Fourth, the recharge coefficient dominates the map through its range, not its magnitude: a 1000×10^3 acre-ft recharge difference moves the head by roughly 17.5 ft at the pre-permit coefficient ($\hat{\beta} \approx 0.0175$; the pumpage coefficient on that train is -0.031 ft per 10^3 acre-ft, larger per unit but acting on a range of ∼300–540 against recharge's 44–2 486), and γ is an ordinary-least-squares association the paper itself diagnoses as simultaneity-contaminated (S17), not a policy lever. The full-sample coefficients $(\hat{\beta}, \hat{\gamma}) = (0.017, -0.026)$ differ from the pre-permit window's because the windows differ.

Table S23: Pumpage counterfactuals, 1991–2023 (fitted pre-permit map; actual recharge). Pumpage: 1991–2023 mean in 10^3 acre-ft yr^{-1}; heads, minimum head over the path and RMSE against the observed record in ft. The observed 2023 head is 635.7 ft.

Scenario	pumpage	head 2023	min. head	RMSE
actual pumpage	382.1	641.6	640.2	8.56
pumpage frozen at 1990	489.4	630.9	630.5	14.22
pre-permit mean pumpage	469.8	632.4	632.0	13.02
20% pumpage cut	305.7	646.8	645.5	**7.19**

S22. Climate-informed recharge: fixed windows, the recharge target, and the Uvalde note

Fig. S7 shows the rolling climate-pass scores of main-text Table 9. Same-year corr(R, precipitation) = 0.78, which is why the precipitation oracle yields 10.56 ft; this uses target-year precipitation unavailable at the origin and is a conditional diagnostic, not a forecast. It remains worse than the realized-flux conditional diagnostic (7.55 ft) because the linear rain map misses 1957-scale extremes. On the 1957–61 recovery no causal climate-informed module has lower RMSE than persistence (persist 43.6 ft; best causal 48.8 ft; precipitation diagnostic 33.7 ft); September–November 1956 is La Niña (−0.92) and does not announce $R_{1957} = 1143$. The remaining fixed windows complete the record: drawdown 1951–56, M2_Renso 24.30, M2_Rprecip 28.67, M2_combo 24.70, M2_Rar 31.38, precipitation diagnostic 18.16 ft; pre-permit wet 1991–95, M2_Rprecip 22.03, M2_Renso 23.03, M2_Rar 23.94, M2_combo 25.71, diagnostic 10.98 ft; critical-period era

2015–23, M2_Rprecip 14.52, M2_Rar 14.67, M2_Renso 16.01, M2_combo 16.57, diagnostic 9.75 ft; on this one window the two climate-informed modules M2_Rprecip and M2_Rar edge past M1 by about one foot (14.52 and 14.67 against 15.62 ft). On the recharge target itself the fixed-window scores are an order of magnitude coarser on every window (climate modules 199–937; precipitation diagnostic 80.7–487, against the 556×10^3 acre-ft climatology scale of the rolling record), so the marginal head advantage is a window-specific result, not a recharge forecast. Additional climate indices (PDO, AMO) were not run; they would enter through the same one-step recharge forecast and cannot alter the timing result. A mid-year data-update forecast would require a new origin definition and evaluation protocol.

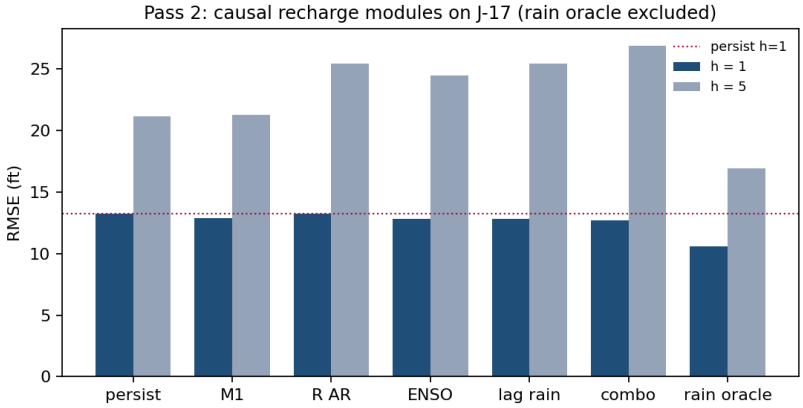


Figure S7: Rolling RMSE on J-17 for climate-informed recharge. The precipitation diagnostic uses year $t + h$ precipitation, unavailable at the origin, and is excluded from retention.

The Uvalde note. This post-freeze exploratory port of the ladder to the separately regulated Uvalde Pool of the same aquifer (index well J-27; annual head lag-one autocorrelation 0.84) finds the AR(1) losing to persistence by 0.13 ft at $h = 1$, consistent with the directional expectation that on a more strongly autocorrelated head record the AR(1)'s edge over persistence should be smaller. A single additional pool is not a test or independent confirmation; no Uvalde result enters any retention verdict, the two records are not pooled, and the comparison is not offered as a test of anything claimed in the main text.

Part C: Cross-system material

S23. Rolling-origin decomposition and baseline horizons of Section 5

Table S24 gives the rolling-origin RMSE of last-value persistence and of the training mean at every horizon $h = 1, \ldots, 5$ on the three scored series, computed from the archived data files on the origin sets of the scored ladders (first origin 1990 with an 8-year minimum training window on Specification A; 1965 with 12 years on Specification B; 1948 with 15 years on J-17), together with the lag-h autocorrelation ρ_h of the predictand over the full series and the stationary prediction $\sqrt{2(1-\rho_h)}$ of main-text Remark 2 for the ratio of the two RMSEs. The $h = 1$ and $h = 5$ values reproduce the archived summaries (main-text Tables 3 and 7) to the printed precision: 98.0/264.7 and 423.6/507.4 kt on Specification A, 84.4/300.0 and 458.1/521.6 kt on Specification B, 13.23/21.11 and 16.17/16.80 ft on J-17. On J-17 the observed ratio tracks the prediction within 0.11 at every horizon, always from below, and crosses one between $h = 1$ and $h = 2$, where ρ_h falls through one half. On both cod series the observed ratio stays far below the prediction at every horizon, the training mean of a collapsing series being a poor forecast of anything; the stationary formula would place a crossover between four and five years on Specification A and beyond five on Specification B, and neither occurs. The script is `work/horizon_profile.py`; its JSON summary is archived beside it.

For the exact rolling-origin application of main-text Proposition 2, at origin i let $(\alpha_i, \beta_i, \gamma_i, \delta_i)$ be the coefficients fitted for M2 using only the training sample. Before output clipping, its increment evaluated at the realized target-year flux is

$$m_i = \alpha_i + \beta_i R_{t+1} + \gamma_i P_{t+1} + \delta_i H_t, \qquad \hat{m}_i = \alpha_i + \beta_i R_t + \gamma_i P_t + \delta_i H_t,$$

where $\hat{m}_i$ is the increment used by the forecast, with fluxes persisted from the origin. Define $u_i = \Delta H_{t+1} - m_i$ and $d_i = m_i - \hat{m}_i$. The identity is checked origin by origin as

$$(\Delta H_{t+1} - \hat{m}_i)^2 - (\Delta H_{t+1})^2 = (d_i^2 - m_i^2) - 2u_i\hat{m}_i.$$

Across all 75 one-year origins, the mean substitution term $\overline{d^2 - m^2}$ is $+76.028642\,\mathrm{ft}^2$, the mean residual cross-term $-2\overline{u\hat{m}}$ is $-27.490666\,\mathrm{ft}^2$, and their sum, the raw M2-minus-persistence MSE gap, is $+48.537976\,\mathrm{ft}^2$. Recomputed origin-specific M2 and persistence forecasts match the

Table S24: Horizon profile of the two baselines on the scored origin sets. RMSE in kt (cod) and ft (J-17); ratio = RMSE(persistence)/RMSE(training mean); ρ_h is the lag-h autocorrelation of the predictand over the full series.

Series	h	n	persist	mean	ratio	ρ_h	$\sqrt{2(1-\rho_h)}$
J-17 head (ft)	1	75	13.23	16.17	0.818	0.629	0.861
	2	74	17.69	16.58	1.067	0.348	1.142
	3	73	20.59	16.88	1.220	0.137	1.314
	4	72	21.50	16.93	1.270	0.061	1.371
	5	71	21.11	16.80	1.256	0.073	1.361
Cod A, SSB (kt)	1	25	98.05	423.59	0.231	0.933	0.365
	2	24	184.32	460.54	0.400	0.825	0.592
	3	23	237.97	488.07	0.488	0.687	0.792
	4	22	256.74	500.18	0.513	0.553	0.945
	5	21	264.72	507.39	0.522	0.406	1.090
Cod B, SSB (kt)	1	59	84.43	458.12	0.184	0.971	0.241
	2	58	152.44	474.51	0.321	0.913	0.417
	3	57	208.89	489.99	0.426	0.838	0.569
	4	56	258.77	505.80	0.512	0.757	0.697
	5	55	299.98	521.56	0.575	0.678	0.802

archived forecasts. After the [610, 710]-ft output clip, one upper clip occurs (origin 1992; raw M2 proposal 719.1027 ft, emitted value 710 ft); no lower clip occurs. The archived scored RMSEs, 14.70 ft for M2 and 13.23 ft for persistence, imply a clipped MSE gap of $+40.988957\,\text{ft}^2$. Thus the decomposition is exact for the un-clipped fitted increments, whereas the output cap changes the scored difference. The implementation and per-origin audit are in `work/edw/increment_identity_rolling.py`, `work/edw/results/increment_identity_rolling.csv` and `work/edw/results/increment_identity_rolling.json`.

Table S25 reports full-series lag-one autocorrelations, means and standard deviations; these are descriptive full-sample diagnostics, not the rolling attribution above. The J-17 flux component is $\hat{\beta}R_t + \hat{\gamma}P_t$ with the full-sample least-squares coefficients of the one-pool map, $(\hat{\alpha}, \hat{\beta}, \hat{\gamma}, \hat{\delta}) = (177.6,\ 0.0173,\ -0.0260,\ -0.273)$; over the 90 annual flux observations its lag-one autocorrelation is 0.078, its mean is 3.52 ft and its standard deviation is 9.20 ft. The apparent net production of the cod series is $S_{t+1} - S_t + C_t$ with

the catch series of the respective pass. Further correlations used in the main text: $r(\Delta H_t, R_t) = 0.742$, $r(\Delta H_t, R_{t-1}) = 0.056$, $r(R_t, R_{t-1}) = 0.172$ as a Pearson correlation of successive pairs (0.170 as the autocovariance-based estimator in the table), and $r(H_t, H_{t-1}) = 0.644$ as a Pearson correlation of pairs (0.629 autocovariance-based; the single-system aquifer study quoted 0.64).

Table S25: Lag-one autocorrelations (autocovariance estimator), means and standard deviations of the series used in main-text Section 5.1. Units: ft and 10^3 acre-ft for J-17; kt for cod.

Series	ρ_1	mean	s.d.
J-17: annual recharge R_t	0.170	682.83	515.44
J-17: annual pumpage P_t	0.870	318.82	114.41
J-17: flux mechanism $\hat{\beta}R_t + \hat{\gamma}P_t$	0.078	3.52	9.20
J-17: head increment ΔH_t	-0.115	-0.39	12.34
J-17: head H_t	0.629	664.42	14.70
Cod A: apparent net production, coarse regime	0.753	57.74	112.71
Cod A: apparent net production, annual landings	0.714	53.00	124.86
Cod B: apparent net production, landings	0.908	184.73	212.65
Cod A: SSB S_t	0.933	300.51	356.51
Cod B: SSB S_t	0.971	512.76	423.25

S24. Verification record for this edition

The verification script `paper11_forecasting_baselines_v68_verification.py` (standard library only) reads the archived data and result files together with the main-text and SI sources and asserts: (a) the carried-forward cod outcome-year, vintage and threshold checks; (b) the printed Edwards tables against the rolling, uncertainty, fixed-window, modern-period and Comal result files; (c) the baseline horizon profiles and full-sample autocorrelations; (d) the 75-origin J-17 increment identity, term means and raw MSE gap, equality with archived h=1 clipped M2 and persistence forecasts, the unique 1992 upper clip, and the clipped gap recomputed from archived squared errors; (e) the full cod power archive, four contiguous batch manifests and hashes, stable seeds, 14,000-row coverage, finite scores, exact two-horizon 5% comparator rule, and raw-to-summary agreement for the per-cell rates, Wilson intervals, hurdle decomposition, rank baseline and Protocol C reference row; and (f) textual invariants for protocol chronology, score-gate/class distinction, conditional realized-flux scope, figure-source provenance and cod interpretation. The archived

verifier output records **237/237 checks passing** on the assembled v68 sources (2026-10-17).